

Week 3 Task Documentation

This document summarizes the theoretical knowledge and practical SOC tasks related to advanced log analysis, threat intelligence integration, incident escalation, alert triage, evidence preservation, and a full SOC workflow simulation.

1. Advanced Log Analysis

Core Concepts:

- **Log Correlation:** Correlating logs from different sources such as firewalls, endpoints, and application logs to detect attack patterns. Example: Linking failed logins (Event ID 4625) with suspicious outbound traffic.
- **Anomaly Detection:** Detecting unusual behavior such as abnormal login times or large data transfers using rule based or statistical analysis.
- **Log Enrichment:** Adding contextual data such as IP geolocation, asset ownership, or user roles to logs for improved investigation.

Objective: Develop the ability to analyze and correlate logs to identify threats and reduce false positives.

2. Threat Intelligence Integration

- **Indicators of Compromise (IOCs):** Malicious IP addresses, hashes, or domains used by attackers.
- **Tactics, Techniques, and Procedures (TTPs):** Methods used by attackers mapped with MITRE ATT&CK.
- **Threat Feeds:** External intelligence sources such as community or commercial feeds used to enrich alerts.
- **Integration in SOC:** Matching detected IPs or hashes against threat intelligence databases.

Objective: Improve detection capability by integrating threat intelligence with SOC monitoring tools.

3. Incident Escalation Workflows

- **SOC Tier Structure:**

Tier 1 analysts perform triage, Tier 2 analysts conduct deeper investigations, and Tier 3 analysts perform advanced threat analysis.

- **Escalation Criteria:** Alerts are escalated based on severity, complexity, and potential business impact.

- **Communication Protocols:** Structured reporting such as Situation Reports (SITREP) and stakeholder communication.

- **Automation:** Use of SOAR tools to automate escalation and response workflows.

Practical Applications

1. Advanced Log Analysis Activities

- Correlate logs such as failed logins with outbound network connections.
- Detect anomalies like high-volume data transfer.
- Enrich logs with geolocation information using plugins.
- Document findings in structured tables.

Timestamp	Event ID	Source IP	Destination IP	Notes
2025.08.18 12:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS request

2. Threat Intelligence Integration Activities

- Import threat intelligence feeds.
- Match alerts with known malicious indicators.
- Enrich alerts with reputation data from threat intelligence platforms.
- Conduct proactive threat hunting using known attacker techniques.

3. Incident Escalation Practice

- Create cases for high priority alerts.
- Escalate incidents to higher SOC tiers.
- Prepare SITREP reports describing the incident and response actions.
- Use automation to assign alerts to responsible analysts.

4. Alert Triage with Threat Intelligence

- Analyze suspicious alerts such as PowerShell execution.
- Validate IP addresses or file hashes with threat intelligence databases.
- Determine alert priority and investigation status.

5. Evidence Preservation and Analysis

- Collect volatile data such as running processes or network connections.
- Acquire memory dumps from compromised systems.
- Maintain chain of custody documentation with timestamps and hashes.

6. Capstone Project: Full SOC Workflow Simulation

- Simulate an attack using a penetration testing framework.

- Detect the attack through SIEM alerts.
- Triage and analyze the alert to confirm malicious activity.
- Contain the attack by isolating affected systems and blocking attacker IPs.
- Escalate the incident to higher tier analysts.
- Prepare a final incident report and management briefing.

Conclusion

This task provides hands-on experience with SOC operations including log analysis, threat intelligence usage, incident response, escalation procedures, and forensic evidence handling. Completing these exercises helps build real-world skills required for SOC analysts and strengthens understanding of security monitoring workflows.